

Logarithmic scalar-field profiles near the Schwarzschild singularity

Marco García
(Dated: July 12, 2026)

We measure the near-singularity logarithmic profile of test scalar fields inside a Schwarzschild black hole in three spatial dimensions. The metric is held fixed, matching the test-field scope of the rigorous Fournodavlos–Sbierski asymptotics, while the scalar retains a nonlinear symmetry-breaking potential. Identical initial data are evolved with a finite-element excision scheme and compared mode by mode with a free scalar. The primary Mexican-hat-to-linear L^2 ratios are 0.923, 0.867, and 0.876 for the finest available $l = 0, 1, 2$ runs, respectively, compared with 1.006 in the spherical oracle. Thus the vacuum structure changes the logarithmic coefficient only within the quantified numerical error budget rather than changing its near-singularity scaling. The $l > 0$ values remain frozen and uncorrected because their truth-window floors, 10.1% and 6.6%, exceed the correction being tested. A profile-dependent window calibration explains 31.0% of the monopole L^2 deficit on common support and leaves 4.16 percentage points as a three-dimensional mesh/extraction systematic; it is diagnostic and does not replace the frozen ratios. Exterior validation gives $M\omega = 0.48772 \pm 0.01244 - i(0.10161 \pm 0.00545)$ from late, large-domain windows, with scatter dominated by the fitting protocol. The calculation therefore supplies a controlled numerical test of kinetic domination in the black-hole interior, explicitly within the Cowling limit.

I. INTRODUCTION

For a massless scalar on a fixed Schwarzschild space-time, Fournodavlos and Sbierski proved that smooth solutions approaching the spacelike singularity admit the expansion

$$\phi(t, r, \Omega) = A(t, \Omega) \ln r + B(t, \Omega) + \mathcal{O}(r|\ln r|), \quad (1)$$

and that logarithmic blow-up is generic in precise energy topologies [1]. The coefficient A is a function on the singular hypersurface rather than a single global amplitude. Its spherical harmonic components are the observables $a_l(t)$ studied below.

The theorem is linear, but the approach to a space-like singularity is often associated with velocity or kinetic domination. In self-gravitating Einstein–scalar systems, monotone, quiescent singular behavior is stable in broad near-FLRW and subcritical Kasner regimes [2, 3]. This motivates a test-field question with a direct numerical answer: does a nonlinear vacuum potential retain an observable imprint in $A(t, \Omega)$, or is that structure washed out as the kinetic terms dominate?

Previous numerical studies provide complementary pieces but not this measurement. Thuestad, Khanna, and Price evolved scalar fields through the interiors of Schwarzschild and Kerr black holes and found interior l -dependent oscillations, without extracting the $r \rightarrow 0$ logarithmic coefficient [4]. Symmetry-breaking scalars have also been evolved in the exterior: accretion can delay symmetry breaking near the horizon [5] or form symmetry-restored shells and trigger vacuum decay [6]. Black-hole-seeded vacuum decay itself has a substantial literature, with both catalytic scenarios and strong counterarguments [7, 8]. These works motivate the physics while leaving the near-singularity profile open.

We present, to our knowledge, the first numerical measurement of $A(t, \Omega)$ approaching the Schwarzschild singularity and the first identical-data comparison of its

free and symmetry-breaking evolutions. The measurement is performed mode by mode in $3 + 1$ dimensions. The methodology is deliberately not claimed as the first scalar evolution with excision: three-dimensional finite-difference excision predates this work [9], and finite elements have been used for black-hole perturbations [10]. The methodological contribution is the combination of a DOLFINx finite-element evolution, calibrated multi-radius interior extraction, paired potential comparison, and an exterior validation that separates finite-domain and overtone systematics.

Section II defines the physical problem and its scope. Section III describes the evolution and observables. Sections IV and V give the interior and exterior results, followed by limitations and interpretation in Sec. VI.

II. PHYSICAL SETUP AND SCOPE

We evolve a real scalar on a Schwarzschild background in ingoing Kerr–Schild coordinates. With signature $(-, +, +, +)$,

$$g_{\mu\nu} = \eta_{\mu\nu} + 2Hk_\mu k_\nu, \quad H = \frac{M}{r}, \quad k_\mu = (1, x_i/r), \quad (2)$$

which is regular at the event horizon. Geometrized units $G = c = M = 1$ are used unless M is shown explicitly.

The scalar action and equation of motion are

$$S_\phi = - \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + V(\phi) \right], \quad (3)$$

$$\square_g \phi - V'(\phi) = 0. \quad (4)$$

Two models are compared. The linear member has $V_{\text{lin}} = 0$. The symmetry-breaking member has

$$V_{\text{hat}}(\phi) = \frac{\lambda}{4} (\phi^2 - v^2)^2. \quad (5)$$

We use $\lambda = 0.10$ and $v = 1.0$, and initialize the nonlinear field in the vacuum approached at the outer boundary. The two members of every comparison receive the same perturbation.

The production perturbation is an ingoing curved Gaussian with amplitude 0.10, center $r_0 = 5.0M$, and width $w = 1.0M$, multiplied by a real Y_{l0} . We evolve $l = 0, 1, 2$. The angular decomposition is performed numerically in three dimensions; no spherical symmetry is imposed on the evolution.

The metric is not evolved. This Cowling or test-field limit is a physical restriction of the study, not an omitted convergence check. It is also the precise background setting of Eq. (1). Our result tests the effect of $V(\phi)$ on the scalar asymptotics; it does not address backreaction, mass inflation, or singularity resolution.

III. NUMERICAL METHOD AND OBSERVABLES

A. First-order finite-element evolution

Writing the metric in $3+1$ form with lapse α , shift β^i , spatial metric γ_{ij} , and extrinsic-curvature trace K , we define

$$\Pi = \alpha^{-1}(\partial_t \phi - \beta^i \partial_i \phi). \quad (6)$$

The evolved system is

$$\partial_t \phi = \alpha \Pi + \beta^i \partial_i \phi, \quad (7)$$

$$\partial_t \Pi = \alpha D_i D^i \phi + D^i \alpha D_i \phi + \beta^i \partial_i \Pi + \alpha K \Pi - \alpha V'(\phi). \quad (8)$$

We discretize both variables with continuous piecewise-linear elements on a graded tetrahedral Gmsh mesh [11] and assemble the weak form in DOLFINx [12]. Time integration uses SSP-RK3. The black-hole interior is excised inside the horizon, where no incoming characteristic requires inner boundary data. At the outer boundary we use a characteristic radiative condition; exterior spectroscopy additionally uses a smooth damping layer.

We record the Killing-energy change together with the inner and outer fluxes as an implementation diagnostic. Its production residual is dominated by surface quadrature on the faceted excision boundary, so it is not promoted as a field-accuracy metric; convergence of the evolved field and the paired profile observables is assessed separately.

The linear operator is preassembled. Only the cubic part of $V'_{\text{hat}} = \lambda \phi(\phi^2 - v^2)$ is assembled at each stage. The consistent mass matrix is retained in production. A direct A/B test found that row-sum mass lumping displaced the interior observable by 27.7% in the median and 38.6% at maximum. It was therefore rejected despite its computational speed.

B. Near-singularity estimator

Following the full hierarchy associated with Eq. (1), the primary radial model for each multipole is

$$\phi_l(t, r) \simeq a_l(t) \ln r + b_l(t) + c_l(t) r \ln r + d_l(t) r. \quad (9)$$

The fit is performed over $r/M \in [0.10, 0.50]$ with truncation order 1. A lower-order fit over the same radii supplies only the phase anchor. The primary order is a bias-variance compromise: higher orders reduce the one-dimensional truncation bias but amplify correlated three-dimensional mesh errors.

Statistics use the anchored strong phase $t/M \in [4.0, 10.0]$ and require the anchor magnitude to exceed 0.30 of its maximum in that interval. This upper time cut removes late finite-domain contamination. For one-dimensional truth we use the same asymptotic order on a deeper interval, $r/M \in [0.02, 0.20]$ at order 1.

C. Paired discriminator and calibration

The primary identical-data discriminator for each l is

$$\mathcal{D}_l = \frac{\|a_{\text{hat},l}\|_{2,\mathcal{S}}}{\|a_{\text{lin},l}\|_{2,\mathcal{S}}}, \quad (10)$$

where $\mathcal{S}$ is the common strong support. The median and interquartile range of $a_{\text{hat},l}/a_{\text{lin},l}$ are secondary summaries. Pointwise ratios near a zero crossing are ill-conditioned. Ratios of separate maxima are excluded because the maxima occur at different times and inherit a differential window bias.

For $l = 0$, a diagnostic calibration is obtained by applying the production fit and the deeper truth fit to the same dense oracle snapshot,

$$c_X(t) = \frac{a_{X,\text{prod}}(t)}{a_{X,\text{truth}}(t)}. \quad (11)$$

The ratio is defined only in the oracle strong phase and is never filled across its zero-crossing gap or beyond its support. Corrections are applied to both members of a pair before recomputing a diagnostic discriminator. They do not replace the frozen production values.

IV. INTERIOR RESULTS

A. Logarithmic profiles

Figure 1 compares the three-dimensional monopole profiles with the dense spherical oracle and Eq. (9). The curves are approximately affine in $\ln r$, and the same basis describes both potentials across representative parts of the common strong phase.

At the finest production rung, the median three-dimensional-to-oracle deviation is 13.1% for the linear

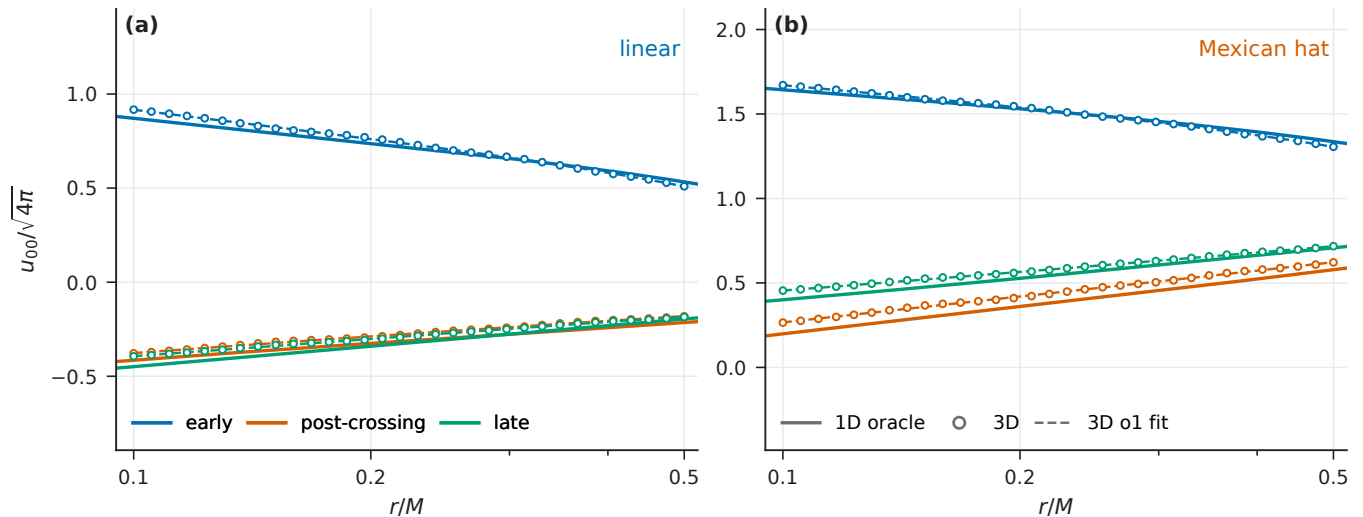


FIG. 1. Near-singularity monopole profiles for identical initial data in the linear and Mexican-hat models. Solid curves show the dense spherical oracle, open circles the versioned low-resolution 3 + 1D profile bank, and dashed curves the primary order-one fits. Colors select reproducible representatives of the common strong phase. This is a qualitative low-resolution profile comparison; quantitative deviations and the L^2 discriminator are reported separately.

TABLE I. Frozen L^2 discriminator at the finest available rung. The $l > 0$ values are not calibration-corrected; their truth-window floors are stated in the text.

l	$\mathcal{D}_l$	median [IQR]
0	0.923	0.851 [0.574, 1.175]
1	0.867	0.640 [0.329, 0.884]
2	0.876	0.813 [0.488, 0.975]

field and 8.6% for the Mexican-hat field. These comparisons include temporal interpolation and extraction floors; they are not pure mesh uncertainties.

B. Identical-data discriminator

The mode-by-mode result is summarized in Table I and Fig. 2. In every mode the nonlinear coefficient remains of the same order as its linear partner. This is the operational signature of vacuum-structure erasure: the potential changes amplitudes within the numerical budget but does not introduce a distinct near-singularity scaling.

For $l = 1$ and $l = 2$, the truth-window floors are 10.1% and 6.6%, respectively. They are comparable to or larger than the effect one would attempt to correct, so promoting either an intrinsic or transferred $l > 0$ correction would add unjustified precision. The fine-rung RMS differences for the individual monopole series are 9.3% and 11.8% for the linear and nonlinear runs. Their paired discriminator is visibly more stable than either absolute comparison because the members share the mesh, estimator, and initial data.

C. Profile-dependent window systematic

On their oracle strong phases, the sampled calibration factors have medians 1.020 [0.982, 1.042] and 1.014 [0.993, 1.069] for the linear and Mexican-hat profiles. Figure 3 displays their time dependence and the strict-support deviation budget.

For the fine linear run the common-support median deviation changes from 10.31% to 7.11%, accounting for 31.0% of that median deviation. The corresponding Mexican-hat values are 4.83%, 3.89%, and 19.3%. These are systematic diagnostics on reduced support, not substitutes for the production values quoted above.

D. Off-channel response

Nonlinear mode coupling is constrained by comparing each off-channel signal with its linear control over the same capped strong phase. For the fine $l = 2$ pair, the $l = 0$ off-channel fractions are 1.95% (linear) and 1.89% (nonlinear), while the $l = 4$ fractions are 3.38% and 3.40%. The pairs are indistinguishable at this level. The $l = 1 \rightarrow 0$ pair changes from 1.61% to 3.50% and is the only weak excess. Only an excess over the paired linear junk constrains coupling; no isolated fraction is treated as a detection.

V. EXTERIOR VALIDATION

We validate the same three-dimensional evolution and extraction stack using the fundamental scalar

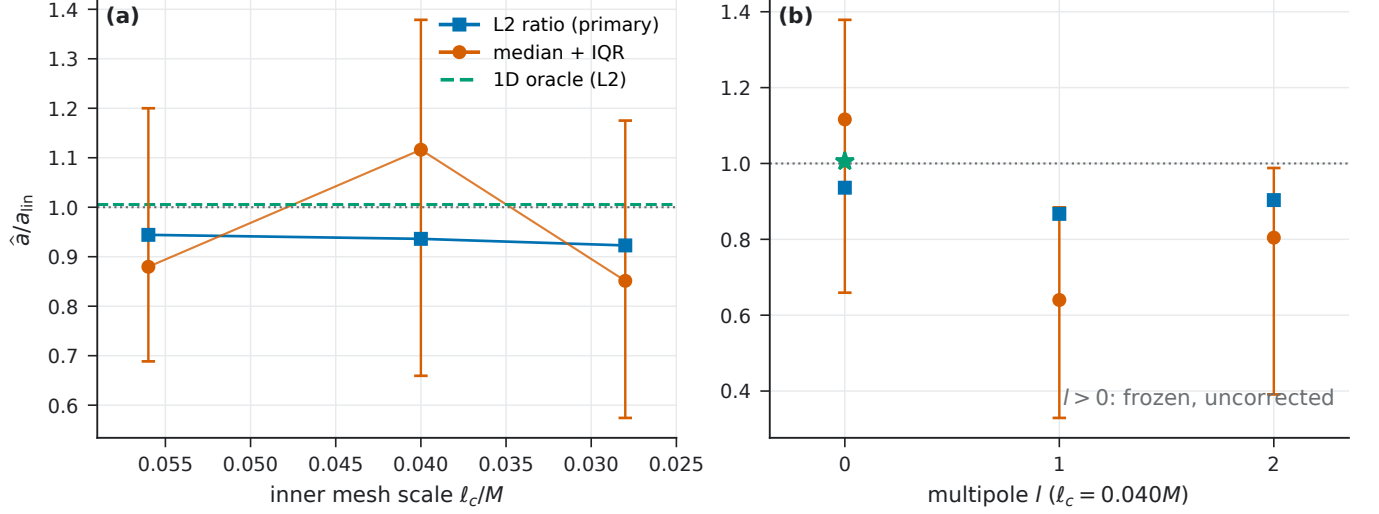


FIG. 2. Identical-data Mexican-hat-to-linear discriminator. Squares show the frozen L^2 ratio, the primary statistic; circles and bars show the secondary median and IQR. The dashed line is the spherical-oracle L^2 ratio. The $l > 0$ values are frozen and uncorrected because their truth-window floors exceed the correction being tested. Peak ratios and corrected common-support L^2 diagnostics are deliberately omitted.

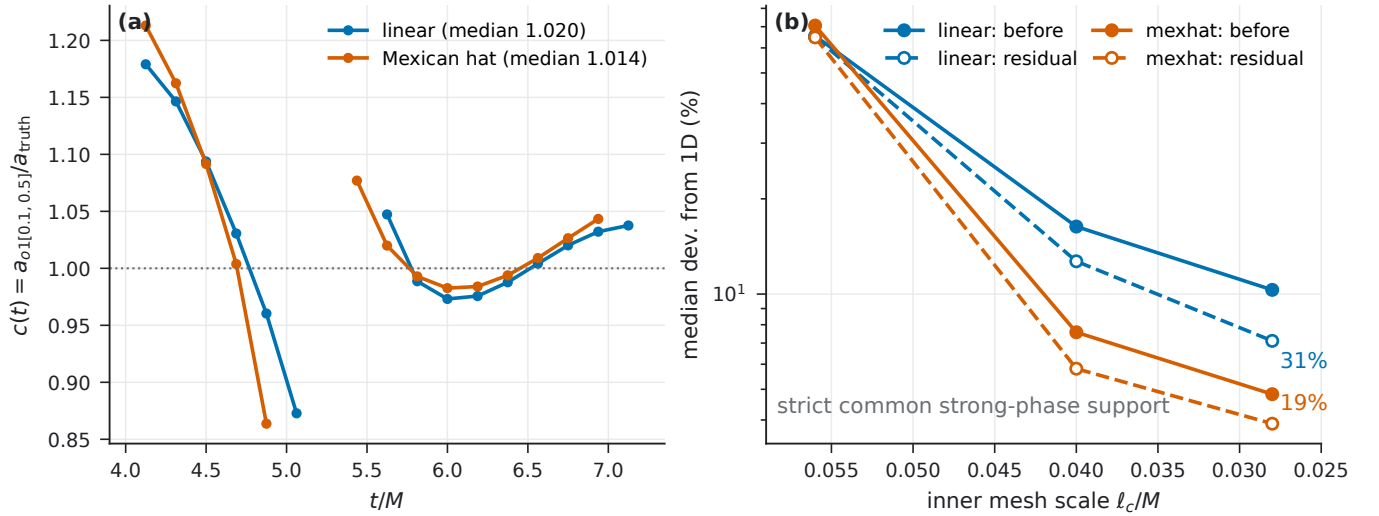


FIG. 3. Window calibration of the monopole estimator. Left: $c(t)$ for the linear and Mexican-hat oracles, defined only on strong-phase support and left unfilled elsewhere. Right: median 3D–1D deviation before and after the diagnostic calibration on strict common support. Residual curves do not replace the frozen production deviations; the coarsest rung is shown diagnostically and is not used for a convergence claim.

$l = 2$ Schwarzschild quasinormal mode. The reference frequency is obtained independently by the Leaver continued-fraction method [13]. Its value is $M\omega_L = 0.48364 - i0.09676$.

We excite $(l, m) = (2, 0)$ and extract at $r = 6.0M$. The exterior pulse has amplitude 0.001, center $8.0M$, and width $2.0M$. The designed fit uses peak-anchored windows of length $26.0M$, begins its peak search at $12.0M$, and permits 4 Prony modes.

A. Real frequency and damping

Figure 4 separates two effects that a single window average would mix: mesh behavior of the real frequency and first-overtone bias of the damping rate.

At the fine small-domain rung, $\text{Re}(M\omega) = 0.47455$ with signed error -1.88% relative to Leaver. The fine-pair order is $p = 1.77$; a three-point Richardson extrapolation is not used because the coarse rung lies outside a monotone asymptotic regime.

Pooling declared late windows from two matched large-

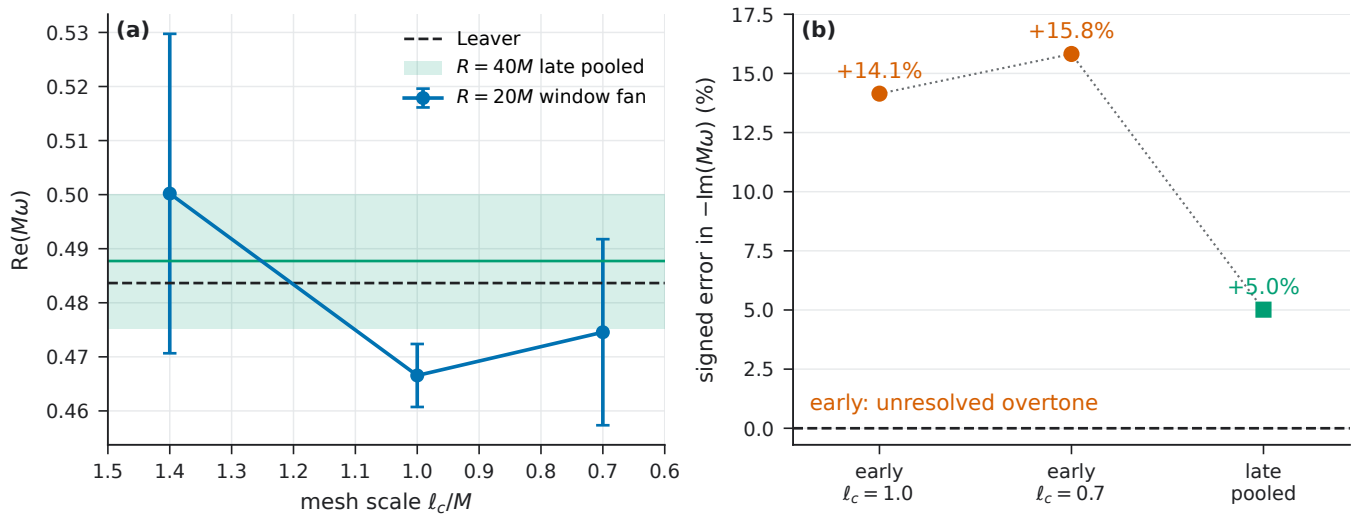


FIG. 4. Exterior $l = 2$ spectroscopy. Left: real-frequency ladder from anchored window fans, with the Leaver value and the late pooled large-domain band. Error bars and the band represent fit/window scatter, not pure discretization uncertainty. Right: early-window damping biases from the unresolved first overtone compared with the late pooled estimate. The early points are protocol biases, not fundamental-mode measurements; the retracted small-domain damping estimate is not shown.

domain rungs gives

$$\text{Re}(M\omega) = 0.48772 \pm 0.01244, \quad (12)$$

$$-\text{Im}(M\omega) = 0.10161 \pm 0.00545. \quad (13)$$

The signed errors are +0.84% and +5.01%. These uncertainties are scatter across late windows and two rungs and are dominated by the fitting protocol. Early large-domain windows instead overestimate the damping by 14.1% and 15.8%. The first overtone is not separable in the adopted finite window, so those values quantify a protocol bias rather than alternative measurements of the fundamental. This controlled scalar example mirrors the broader ringdown start-time and overtone debate [14, 15].

B. Finite-domain cavity

The small exterior domain develops a late trapped signal between the black-hole potential barrier and the damping layer. A matched large domain preserves the inner mesh grading while moving the layer outward, isolating this domain effect.

Figure 5 shows the paired waveform and tail spectrum. The matched enlargement reduces the tail floor by a factor 47.1 and raises peak-to-floor from 4.1 to 195.5. The small-domain lines are $M\omega_1 = 0.3535 \pm 0.0010$ and $M\omega_2 = 0.5630 \pm 0.0040$. They are trapped-domain diagnostics, not physical quasinormal frequencies. The matched-domain real-frequency shift, $\Delta\text{Re}(M\omega) = -0.00221 \pm 0.00582$, remains within the window scatter.

VI. DISCUSSION

The central result is not equality of the two potentials to arbitrary precision. It is the persistence of the same logarithmic asymptotic coefficient at order unity for identical data, in all measured multipoles, with a separately measured error budget. Absolute three-dimensional-to-one-dimensional deviations are larger than the rung-to-rung spread of the paired discriminator because correlated mesh and estimator errors cancel to first order in the pair.

The monopole calibration classifies the window bias as a partial explanation: it accounts for 31.0% of the L^2 deficit on common support and leaves 4.16 percentage points. The residual is therefore carried as a genuine three-dimensional mesh/extraction systematic. We do not use the corrected diagnostic as a replacement result, and we do not transfer it to $l > 0$.

This behavior is complementary to exterior symmetry-restoration studies [5, 6]. Near the horizon, redshift and accretion can delay or reorganize vacuum dynamics; deeper inside, our fixed-background calculation shows that the coefficient controlling the singular profile is instead dominated by the common kinetic evolution. Modified-gravity scenarios can produce qualitatively different conclusions. For example, Bars finds Higgs-controlled extensions through the singularity in a locally conformal framework with antigravity sectors [16]. That contrast does not compete with the present result: our claim is explicitly restricted to a minimally coupled scalar on a fixed GR background.

The main limitations are accordingly clear. Backreaction is absent; the calculation does not decide the geometry of a self-gravitating singularity. The radial fit covers

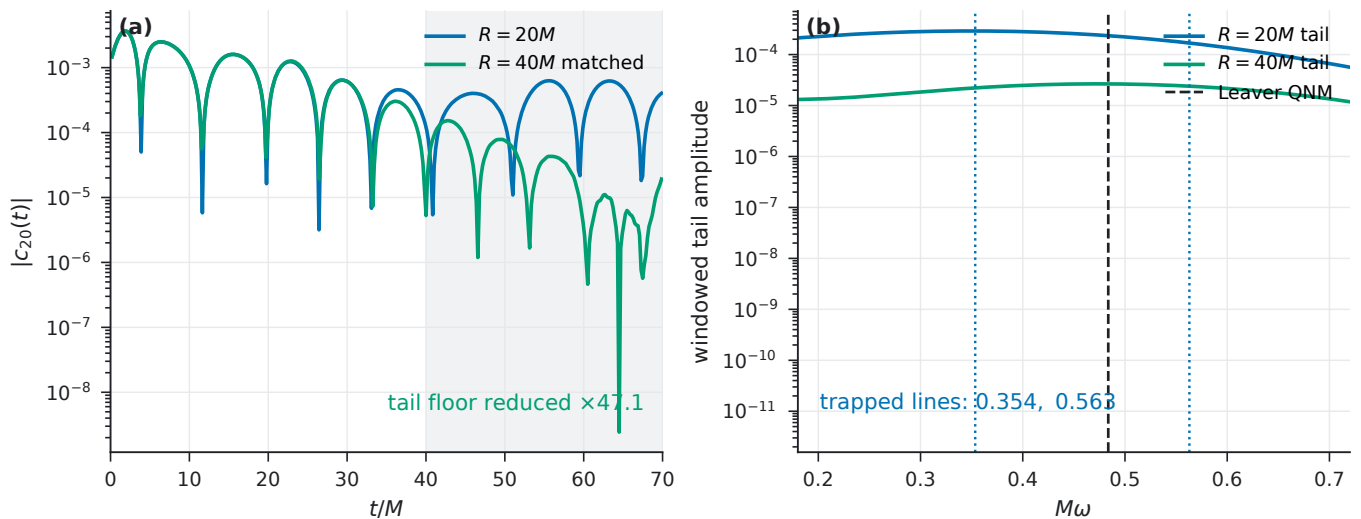


FIG. 5. Paired-domain waveform and tail spectrum. Enlarging the matched domain suppresses the finite-domain tail floor, while the small-domain spectrum contains two persistent trapped lines. The marked lines are finite-domain cavity modes, not physical QNMs; the Leaver frequency is shown only as the physical reference.

a finite interval rather than the mathematical limit. The $l > 0$ truth-window floors prevent a citable profile correction. The interior strong phase ends before the late quasi-stationary regime, and the exterior damping uncertainty remains protocol dominated. Finally, the l -dependent oscillation cross-check suggested by Ref. [4] would require promoting additional versioned $l > 0$ profile series and is deferred.

VII. CONCLUSIONS

We have extracted the Fournodavlos–Sbierski logarithmic coefficient from three-dimensional finite-element evo-

lutions inside Schwarzschild and compared a free scalar with a symmetry-breaking scalar using identical data. The frozen primary ratios 0.923, 0.867, and 0.876 show that the nonlinear vacuum structure does not generate a parametrically different near-singularity profile in $l = 0, 1, 2$. A conservative calibration isolates part of the monopole window bias without promoting a correction beyond its support. Independent exterior spectroscopy reproduces the Schwarzschild QNM within a protocol-dominated uncertainty and exposes both the unresolved-overtone bias and a finite-domain cavity floor. Together, these results support kinetic-domination erasure of vacuum structure in the test-field Schwarzschild interior while making the Cowling limit and all resolved systematics explicit.

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